Usage Guide

FSP Example Project Usage Guide

User's Manual

Renesas RZ Family

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Rev.1.01 Mar 2026

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(Rev.4.0-1 November 2017)

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Renesas RZ Family

FSP Example Project Usage Guide

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1. Introduction

This Example Project Usage Guide provides steps and guidelines for operating projects which use API and source generated by Renesas Flexible Software Package **(FSP)**.

1.1 Prerequisites

1. Tool experience: It is assumed that the user has prior experience working with integrated development environments, such as e^2 studio, SEGGER J-Link RTT Viewer, and terminal emulation programs such as Tera Term.
2. Subject knowledge: It is assumed that the user has basic knowledge about microprocessors, embedded systems, and FSP to modify the example projects. First time users are recommended to refer to FSP User Manual for Tutorial on **Getting Started with Flexible Software Package**, paying special attention to sections as follows.
    - Set up Evaluation Board
    - Tutorial: Your First RZ MPU Project – Blinky
    - Importing an Existing Project into e2 studio

   Prior to running the example projects or programming the kits, all jumper settings must be set to the default settings. Refer to each kits' user's manual for the default jumper settings. After default jumper settings have been restored, refer to the readme.txt for each example project to modify the values from the default.

3. The screen shots provided throughout this document are for reference. The actual screen content may differ depending on the version of software and development tools used.

2. Hardware and Software Requirements

RZ FSP Example projects are designed to operate using RZ MPU kits officially supported by Renesas. Supported kits are identified for each group of microprocessors on the Renesas website.
Refer to the readme.txt file in the specific module folder of */example_projects* folder for additional hardware and software requirements for running the projects.

Note:

1. Some projects may require external hardware as mentioned in the respective readme.txt files
2. Some pin numbers may be printed on the back of the Evaluation Kit board.

Software Requirements

- Windows® 11 operating system

Below software requirements are provided as reference. Please refer to the Release Notes & Tools requirement for the FSP version of your choice to identify exact versions of software required.

- e² studio v2025-12 or later
- RZ FSP v4.0.0 or later
- Arm GNU Toolchain for Cortex-M and Cortex-R: 13.3.Rel1
- Arm GNU Toolchain for Cortex-A: 13.2.Rel1
- SEGGER J-Link RTT Viewer v8.60 or later
- SEGGER J-Link Software for CM33 cold boot on RZ/V2H: 7.96e

3. Tool Installation

3.1 FSP and tools installation

Download and install the latest version of FSP and tools from FSP GitHub repository.

1. Open FSP GitHub repository: https://github.com/renesas/rz-fsp
2. Go to the ***Releases*** section of Git and navigate to the desired FSP version.
3. Follow the instructions on installing and using FSP and e^2 studio.

3.2 J-Link RTT Viewer Installation

This section is designated for example projects of RZ/G, RZ/V, RZ/T (CR52), RZ/N (CR52) series as they support J-Link RTT Viewer as terminal console application.

Download and install SEGGER J-Link Software for Windows from https://www.segger.com/downloads/jlink#J-LinkSoftwareAndDocumentationPack.

Default download path is `C:\Program Files\SEGGER\JLink`.

Note: Select version 8.60 or later from the drop-down menu in Version tab.

3.3 Tera Term Installation

This section is designated for example projects of RZ/A, RZ/T (CA55), RZ/N (CA55) series as they support Tera Term as terminal console application.

Go to the Tera Term GitHub Releases page for the latest version. scroll to the Assets section of the latest release. Then download teraterm-<version>-installer.exe

Default installation path is `C:\Program Files (x86)\teraterm`.

4. Importing and Running the Project

4.1 Downloading the Project

4.1.1 Downloading the Project from GitHub

1. Open RZ FSP Examples Repository: https://github.com/renesas/rz-fsp-examples/releases.
2. Navigate to **Assets** section.
3. Download the project bundle for specific kit.

4.1.2 Downloading the Project from Renesas.com.

1. Download the example project bundle for the specific kit from https://www.renesas.com/en/software-tool/rz-flexible-software-package-fsp#downloads

Note: Some example projects for a kit bundle may require an older version of FSP to be installed due to identified problems found in operating with the latest version of FSP.

4.2 Running the Project

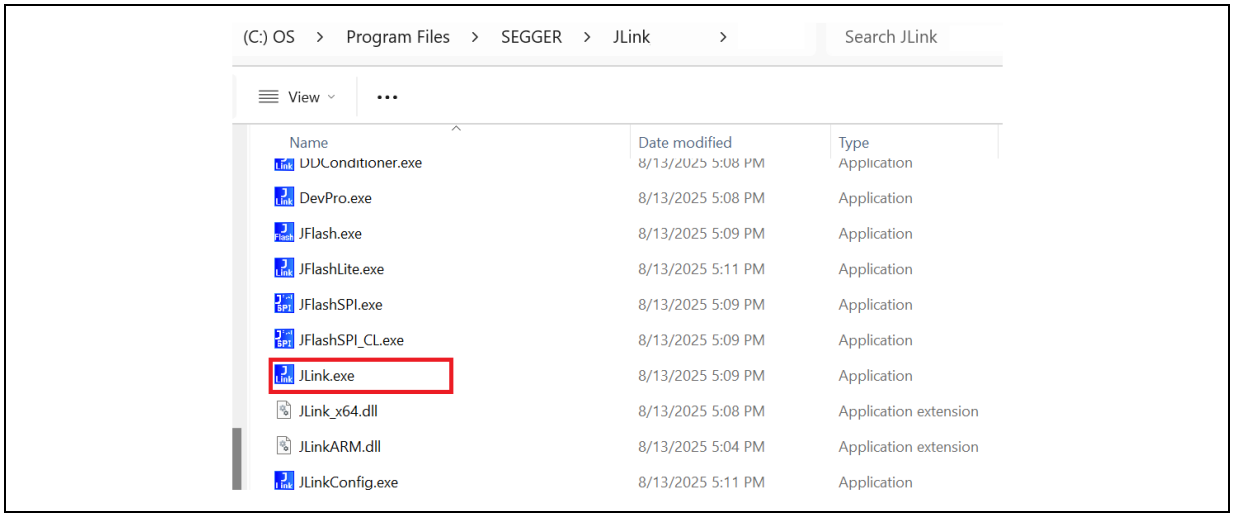
There are two ways of running the project:

- Flashing the pre-built binary (.hex file) and running the project as explained in section 4.2.1.
- Importing the project into e2 studio, building, loading and running the project as explained in section 4.2.2.

4.2.1 Flashing pre-built binary

Follow the steps below to flash the pre-built binary file (.hex). For MCUs with an Arm® Cortex®-A55 CPU core, such as RZ/A3M and RZ/A3UL, initial hardware setup may be required in advance using the RZ/A Initial Program Loader (IPL). In this case, it is recommended to import the project into e² studio as described in Section 4.2.2.

1. In the **e2studio** folder for each module, a prebuilt binary i.e. **.hex** file included.
2. Navigate to the latest downloaded (or installed) `JLink` folder and open SEGGER J-Link Commander by double clicking on `JLink.exe`.



3. In the Command Prompt, type “**connect**” to establish a target connection.
4. Press **Enter** to select the default device option. In case J-Link Commander cannot detect connected device, please type ‘**?**’ for the list of supported devices and choose Renesas RZ device.

```
C:\Program Files\SEGGER\JLir
SEGGER J-Link Commander       (Compiled Aug 13 2025 12:09:33)
DLL version      , compiled Aug 13 2025 12:08:34

Connecting to J-Link via USB...O.K.
Firmware: J-Link OB-S124 compiled Aug 13 2025 10:35:02
Hardware version: V1.00
J-Link uptime (since boot): 0d 00h 25m 16s
S/N: 839007276
USB speed mode: Full speed (12 MBit/s)
VTref=3.300V


Type "connect" to establish a target connection, '?' for help
J-Link>connect
Please specify device / core. <Default>: R9A07G084M04
Type '?' for selection dialog
Device>
```

5. Next, type “**S**” to choose the SWD debug interface. If using JTAG interface, enter “**J**”.

```
Type "connect" to establish a target connection, '?' for help
J-Link>connect
Please specify device / core. <Default>: R9A07G063U02GBG
Type '?' for selection dialog
Device>
Please specify target interface:
  J) JTAG (Default)
  S) SWD
  T) cJTAG
TIF>S
Specify target interface speed [kHz]. <Default>: 4000 kHz
Speed>
```

6. Next, press **Enter** to use the default speed of 4000KHz or input specific speed that’s corresponding to the target device.
   After that, confirm the message “Cortex-R52 identified.” or “Cortex-M33 identified.” is displayed based on corresponding target device.

```
Non-secure status: Non-secure
Cache info:
  Inner cache boundary: none
  LoU Uniprocessor: 1
  LoC: 1
  LoU Inner Shareable: 1
I-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
D-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
Memory zones:
  Zone: "Default" Description: Default access mode
  Zone: "AP0" Description: MEM-AP (APB-AP)
  Zone: "AP1" Description: MEM-AP (APB-AP)
  Zone: "AP2" Description: MEM-AP (AXI-AP)
Cortex-R52 identified.
J-Link>
```

7. Download image file to target device and execute program
   7.1. Example project uses **flash boot mode**
   7.1.1. To load the image file, enter below command, replace “path_to_binary_hex_file” by the path of image file (.hex), for example, “C:\Workspace\rz-fsp-examples\example_projects\rzg2ul_evk\gtm\gtm_rzg2ul_evk_ep\e2studio\ gtm_rzg2ul_evk_ep .hex”

```
loadfile “path_to_binary_hex_file”
```

```
C:\Program Files\SEGGER\JLir
AP[0]: APB-AP (IDR: 0x54770002, ADDR: 0x00000000)
AP[1]: AXI-AP (IDR: 0x44770004, ADDR: 0x01000000)
AP[2]: AHB-AP (IDR: 0x84770001, ADDR: 0x02000000)
AP[3]: AHB-AP (IDR: 0x84770001, ADDR: 0x03000000)
Iterating through AP map to find AHB-AP to use
AP[0]: Skipped. Not an AHB-AP
AP[1]: Skipped. Not an AHB-AP
AP[2]: Core found
AP[2]: AHB-AP ROM base: 0xE00FF000
CPUID register: 0x410FD214. Implementer code: 0x41 (ARM)
Feature set: Mainline
Cache: No cache
Found Cortex-M33 r0p4, Little endian.
FPUnit: 8 code (BP) slots and 0 literal slots
Security extension: implemented
Secure debug: enabled
CoreSight components:
ROMTbl[0] @ E00FF000
[0][0]: E000E000 CID B105900D PID 000BBD21 DEVARCH 47702A04 DEVTYPE 00 Cortex-M33
[0][1]: E0001000 CID B105900D PID 000BBD21 DEVARCH 47701A02 DEVTYPE 00 DWT
[0][2]: E0002000 CID B105900D PID 000BBD21 DEVARCH 47701A03 DEVTYPE 00 FPB
[0][3]: E0000000 CID B105900D PID 000BBD21 DEVARCH 47701A01 DEVTYPE 43 ITM
[0][5]: E0041000 CID B105900D PID 002BBD21 DEVARCH 47724A13 DEVTYPE 13 ETM
[0][6]: E0042000 CID B105900D PID 000BBD21 DEVARCH 47701A14 DEVTYPE 14 CSS600-CTI
Memory zones:
  Zone: "Default" Description: Default access mode
Cortex-M33 identified.
J-Link>loadfile C:\Workspace\rz-fsp-examples\example_projects\rzg2ul_evk\gtm\gtm_rzg2ul_evk_ep\e2studio\gtm_rzg2ul_evk_ep.hex
```

When downloading binary image file is successful, the “O.K” status will be displayed as shown in the following figure.

```
Memory zones:
  Zone: "Default" Description: Default access mode
Cortex-M33 identified.
J-Link>loadfile C:\Workspace\rz-fsp-examples\example_projects\rzg2ul_evk\gtm\gtm_rzg2ul_evk_ep\e2studio\gtm_rzg2ul_evk_ep.hex
'loadfile': Performing implicit reset & halt of MCU.
ResetTarget() start
ResetTarget() end - Took 7.49ms
Device specific reset executed.
Downloading file [C:\Workspace\rz-fsp-examples\example_projects\rzg2ul_evk\gtm\gtm_rzg2ul_evk_ep\e2studio\gtm_rzg2ul_evk_ep.hex]...
O.K.
J-Link>
```

7.1.2. Type “**reset**” to reset CPU

```
Cortex-M33 identified.
J-Link>loadfile C:\Workspace\e2_studio\gtm_rzg2ul_evk_ep\e2studio\gtm_rzg2ul_evk_ep.hex
'loadfile': Performing implicit reset & halt of MCU.
ResetTarget() start
ResetTarget() end - Took 9.24ms
Device specific reset executed.
Downloading file [C:\Workspace\e2_studio\gtm_rzg2ul_evk_ep\e2studio\gtm_rzg2ul_evk_ep.hex]...
O.K.
J-Link>reset
Reset delay: 0 ms
ResetTarget() start
ResetTarget() end - Took 13.5ms
Device specific reset executed.
J-Link>
```

7.1.3. Go to section 4.2.3.1 to connect with J-Link RTT Viewer.

7.1.4. Lastly, in Command Prompt window, type “**go**” to start program execution.

```
J-Link>reset
Reset delay: 0 ms
ResetTarget() start
ResetTarget() end - Took 8.66ms
Device specific reset executed.
J-Link>go
Memory map 'after startup completion point' is active
J-Link>
```

7.2. Example project uses **RAM execution without flash memory**

7.2.1. To load the image file, enter below command, replace “path_to_binary_hex_file” by the path of image file (.hex), for example, “C:\Workspace\rz-fsp-examples\example_projects\rzn2l_rsk\spi_dmac\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex“.

loadfile “path_to_binary_hex_file” 0 noreset

```
Processor features:
  EL0 support: AArch32
  EL1 support: AArch32
  EL2 support: AArch32
  EL3 support: N/A
  FPU support: Single + Double + Conversion
Add. info (CPU temp. halted)
Current exception level: EL2
Exception level AArch usage:
  EL0: AArch32
  EL1: AArch32
  EL2: AArch32
  EL3: AArch32
Non-secure status: Non-secure
Cache info:
  Inner cache boundary: none
  LoU Uniprocessor: 1
  LoC: 1
  LoU Inner Shareable: 1
I-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
D-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
Memory zones:
  Zone: "Default" Description: Default access mode
  Zone: "AP0" Description: MEM-AP (APB-AP)
  Zone: "AP1" Description: MEM-AP (APB-AP)
  Zone: "AP2" Description: MEM-AP (AXI-AP)
Cortex-R52 identified.
J-Link>loadfile C:\Workspace\rz-fsp-examples\example_projects\rzn2l_rsk\spi_dmac\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex 0 noreset
```

When downloading binary image file is successful, the “O.K” status will be displayed as shown in the following figure.

```
Non-secure status: Non-secure
Cache info:
  Inner cache boundary: none
  LoU Uniprocessor: 1
  LoC: 1
  LoU Inner Shareable: 1
I-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
D-Cache L1: 16 KB, 64 Sets, 64 Bytes/Line, 4-Way
Memory zones:
  Zone: "Default" Description: Default access mode
  Zone: "AP0" Description: MEM-AP (APB-AP)
  Zone: "AP1" Description: MEM-AP (APB-AP)
  Zone: "AP2" Description: MEM-AP (AXI-AP)
Cortex-R52 identified.
J-Link>loadfile C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex 0 noreset
'loadfile': Skipping reset & halt of MCU before download.
Downloading file [C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex]...
Memory access: CPU temp. halted: https://wiki.segger.com/Memory_accesses#Legacy_stop_mode
O.K.
J-Link>
```

7.2.2. Type below command to set PC at address 0x102000.

```
J-Link>loadfile C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex 0 noreset
'loadfile': Skipping reset & halt of MCU before download.
Downloading file [C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex]...
O.K.
J-Link>setpc 0x102000
J-Link>
```

Note: In case RZ/T2H (Cortex-CA55) and RZ/N2H (Cortex-CA55), use PC address 0x10001000 instead.

7.2.3. Type below command to set CPSR register address to 0x200001DA.

```
J-Link>loadfile C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex 0 noreset
'loadfile': Skipping reset & halt of MCU before download.
Downloading file [C:\Workspace\e2_studio\spi_dmac_rzn2l_rsk_ep\e2studio\spi_dmac_rzn2l_rsk_ep.hex]...
O.K.
J-Link>setpc 0x102000
J-Link>
```

7.2.4. Go to section 4.2.3.1 to connect with J-Link RTT Viewer.

7.2.5. Lastly, in Command Prompt window, type "**go**" to start program execution.

```
J-Link>setpc 0x102000
J-Link>wreg AARCH32_CPSR, 0x200001DA
AARCH32_CPSR = 200001DA
J-Link>go
Memory map 'after startup completion point' is active
J-Link>
```

4.2.2 Importing the project into e2studio

Refer to FSP User Manual for steps on importing a project:

- Importing an existing project
- Generating Project content[1]
- Building the project
- Downloading the project image to the board

Then apply your learnings to the module/project of interest.

Note: In the case of RZ/V2H, it is required to build "preceding_rzv2h_evk_cm33_ep" before building a CR8 project. Hence, please import and build the "..\rzv2h_evk\preceding\preceding_rzv2h_evk_cm33_ep" when trying out the example for CR8.

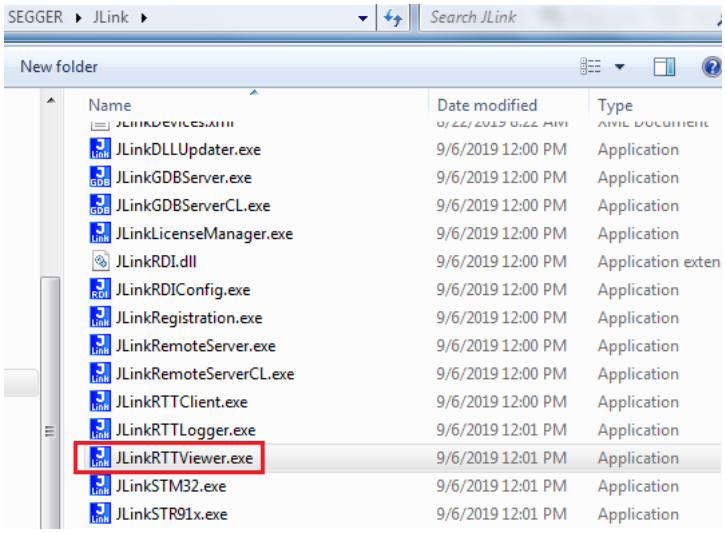
4.2.3 Running the Project

4.2.3.1 Connecting with J-Link RTT Viewer

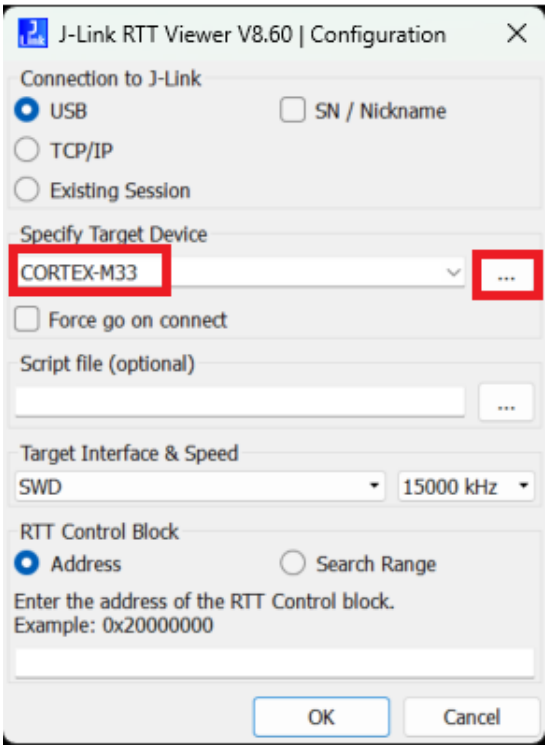
This section is designated for example projects of RZ/G, RZ/V, RZ/T (CR52), RZ/N (CR52) series as they support J-Link RTT Viewer as terminal console application.

[1] Some example projects distributed through https://github.com/renesas/rz-fsp-examples may not be upgraded to the most release of FSP due to technical issues found during regression testing. Refer to the version information table to identify the older version of FSP to use with the non-upgraded example project. When the required FSP version of an example project is not found in the user's installation of tools, the Generate Project Content step will prompt changing the FSP version of the project to the latest version of FSP available locally on the user's PC. It is recommended that the user avoids this and instead opts to download and install the required (and missing) version of FSP and tools. Refer section 0 and 3.1.

1. Open RTT Viewer by double clicking `JLinkRTTViewer.exe` in the downloaded `/Segger/JLink` folder.

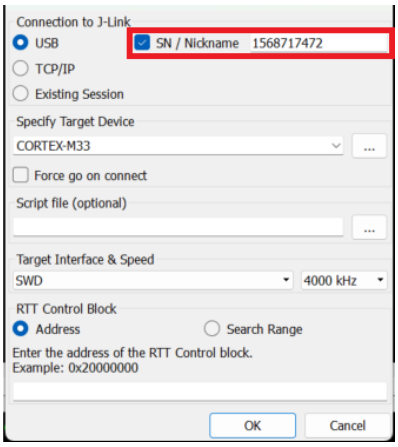


2. On opening, the field **Specify Target Device** shows up as **unspecified**. With USB selected, click on the ... tab and select the desired Renesas RZ device. Keep a note of the Core type.

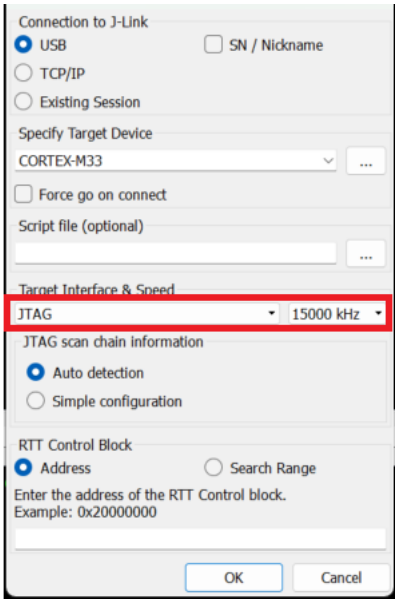


3. If multiple kits are connected to the PC, make sure to choose the corresponding serial number (SN/ Nickname).
   Note: For MPUs, the exact address of the RTT Control Block must be specified. When Auto Detection is used, RTT Viewer parses all RAM memory to find the RTT Block. As TrustZone boundaries are configured for all projects for these MPUs, RTT Viewer will encounter access failures, and you will not see the output logs in RTT viewer. The exact address for the SEGGER_RTT data structure in RAM is

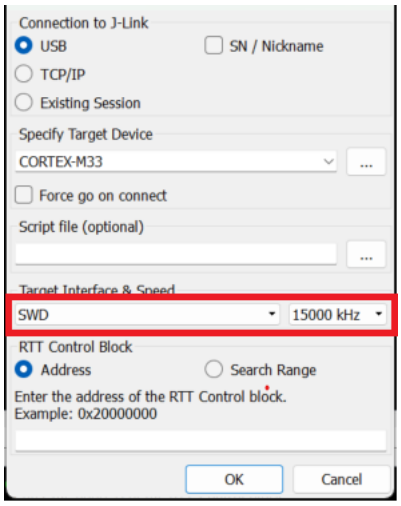
found in the readme.txt file associated with the module under evaluation or in the map file when a binary is built. (Refer Appendix for an example).



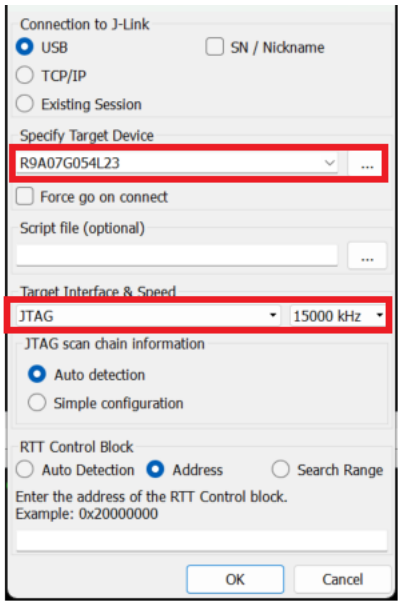
4. Select Target Interface & Speed.
   With respect to the RZ/G2LC, RZ/G2LC, RZ/G2UL examples, please configure them as below.



With respect to the RZ/G3S and RZ/G3E examples, please configure them as below.

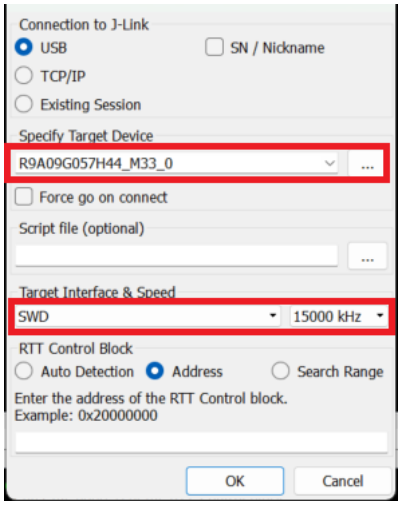


With respect to the RZ/V2L examples, please configure them as below.

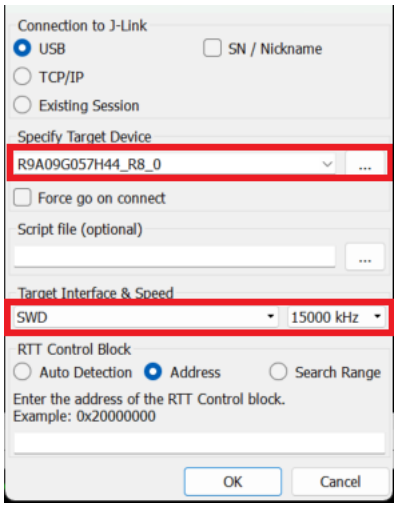


With respect to the other RZ/V2H examples, please configure them as below.

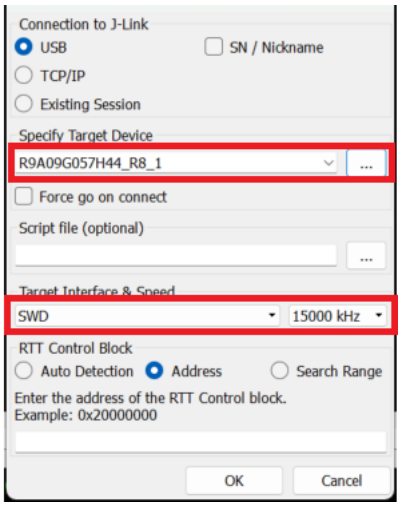
For CM33:



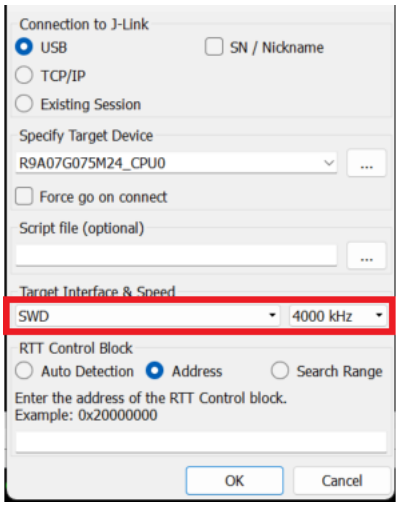
For CR8_0:



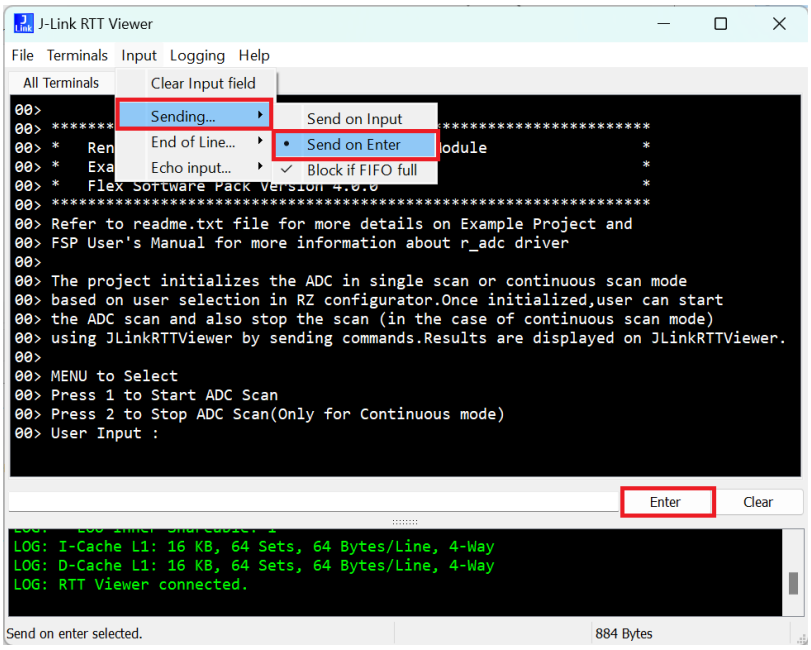
For CR8_1:



With respect to the RZ/T2H (CR52), RZ/T2L, RZ/T2M, RZ/T2ME, RZ/N2H (CR52), RZ/N2L examples, please configure them as below.



5. Click on the **Input** tab and change **Sending** option to **Send on Enter**. Every time input in entered, you must either press the **Enter** or **Enter** tab on the RTT viewer.



6. Follow the instructions displayed on the RTT Viewer as shown above. Also refer to `readme.txt` file in the project folder (downloaded.zip file or in https://github.com/renesas/rz-fsp-examples) to run the project.

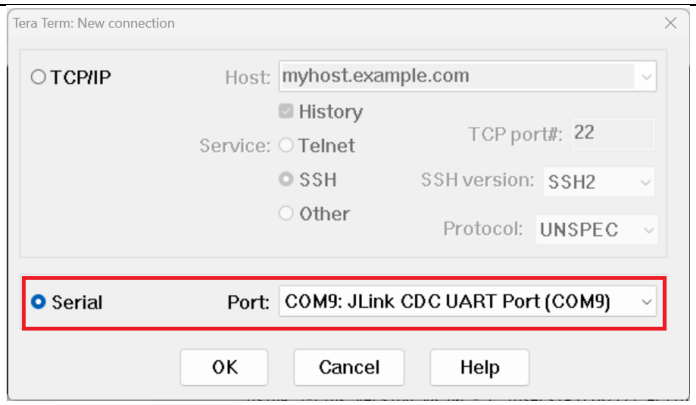
Note:

1. Example Projects do not support floating point or special characters or any non-numeric characters.
2. Example projects do not handle cases where the user's input is greater than the expected input array size.

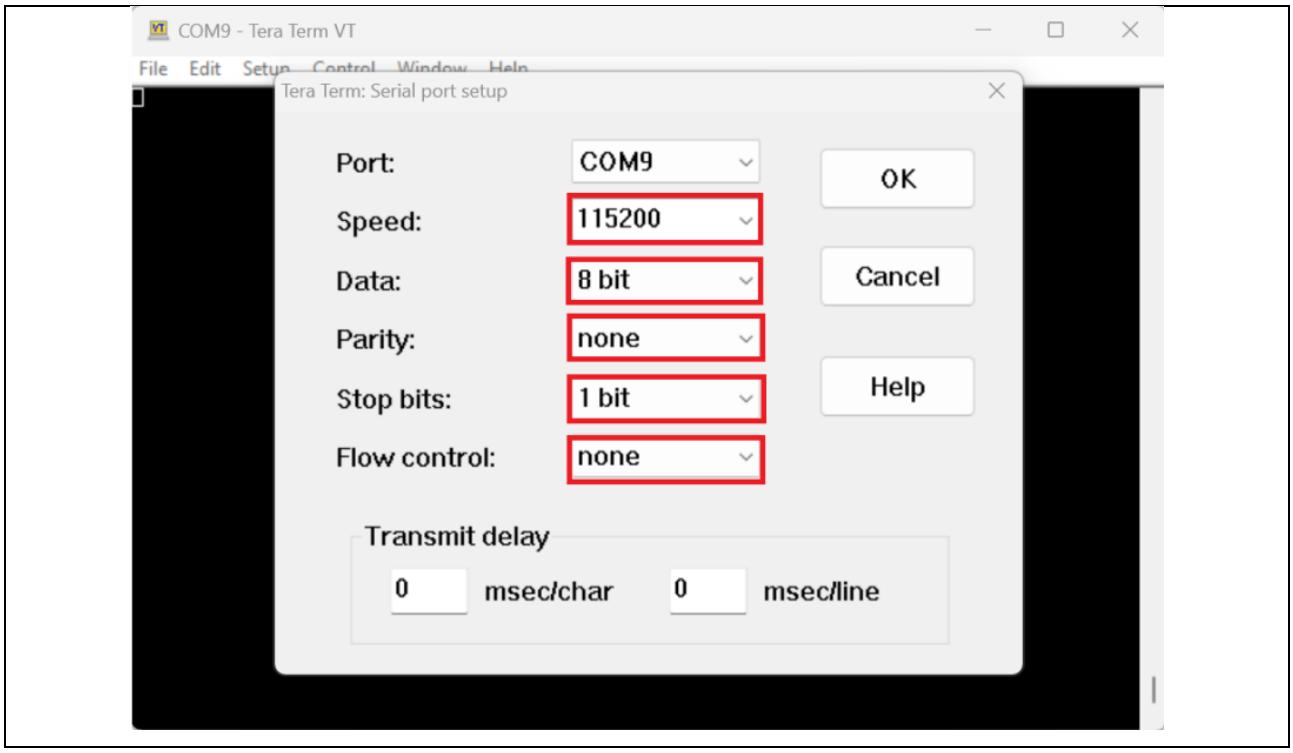
4.2.3.2 Connecting with Tera Term

This section is designated for example projects of RZ/A, RZ/T (CA55), RZ/N (CA55) series as they support Tera Term as terminal console application.

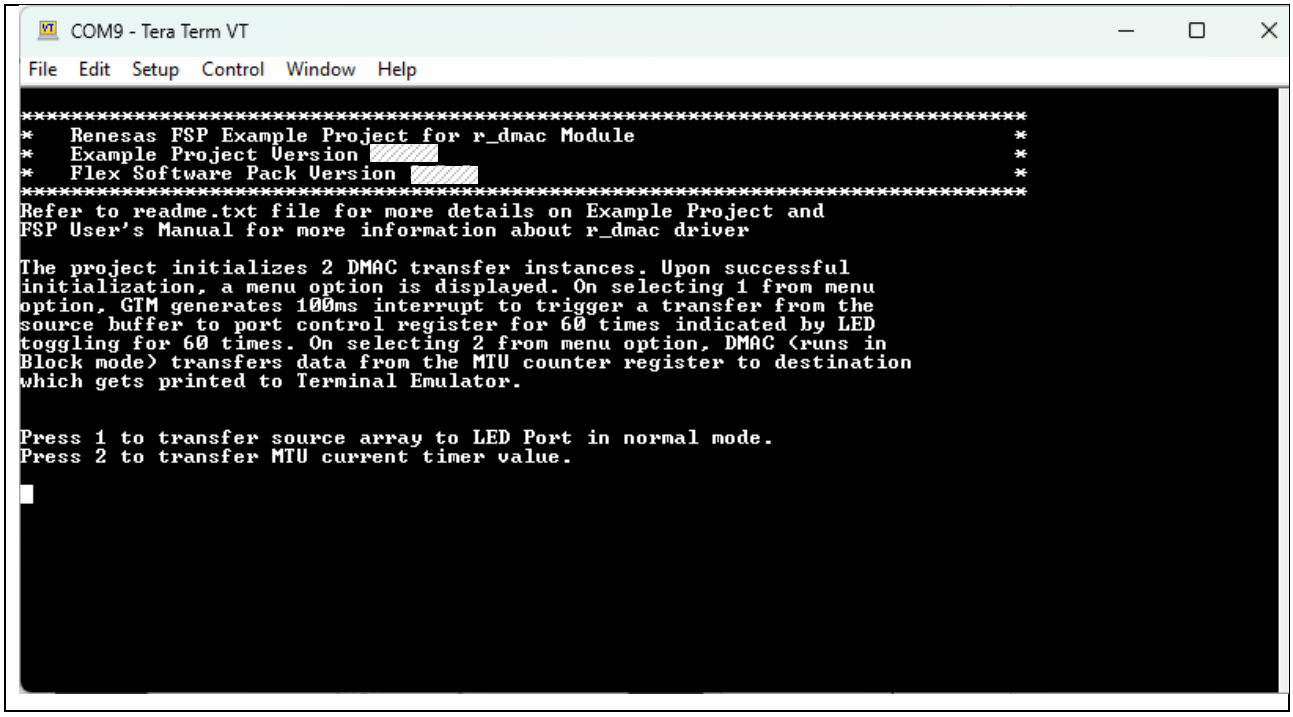
1. Open **Tera Term** by double clicking **ttermpro.exe** in the downloaded folder.
2. Select **Serial port**.



3. Set up **Serial port**.
   With respect to the RZ/A3M, RZ/A3UL, RZ/T2H (CA55), RZ/N2H (CA55) examples, please configure Serial Port as below.



4. Check log in Tera Term.



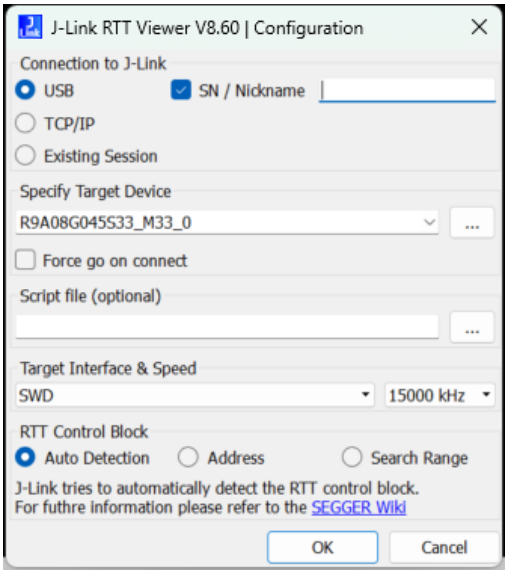
5. References

FSP GitHub:	https://github.com/renesas/rz-fsp
FSP User Manual:	
	www.renesas.com/rza-getting-started-flexible-software-package
	www.renesas.com/rzg-getting-started-flexible-software-package
	www.renesas.com/rzt2-rzn2-getting-started-flexible-software-package
	www.renesas.com/rzv-getting-started-flexible-software-package
	https://renesas.github.io/rz-fsp
FSP Example Projects:	https://github.com/renesas/rz-fsp-examples
Evaluation Kit Manuals:	
	www.renesas.com/rz/ek-rza3m
	www.renesas.com/rz/rza3ul_evk
	www.renesas.com/rz/rzg2l_evk
	www.renesas.com/rz/rzg2lc_evk
	www.renesas.com/rz/rzg2ul_evk
	www.renesas.com/rz/rzg3e_evk
	www.renesas.com/rz/rzg3s_evk
	www.renesas.com/rz/rzn2h_evk
	www.renesas.com/rz/rzn2l_rsk
	www.renesas.com/rz/rzt2h_evk
	www.renesas.com/rz/rzt2l_rsk
	www.renesas.com/rz/rzt2m_rsk
	www.renesas.com/rz/rzt2me_rsk
	www.renesas.com/rz/rzv2h_evk
	www.renesas.com/rz/rzv2l_evk
	www.renesas.com/rz/rzv2n_evk
Knowledge Base:	Knowledge Base | Renesas Customer Hub
Support:	Renesas Engineering Community

6. Appendix

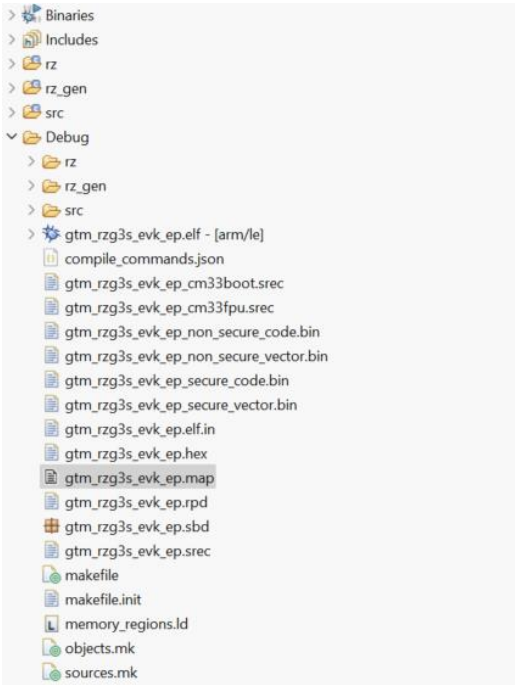
6.1 Limitations in connecting with J-Link RTT Viewer v8.60 or later

When using Auto Detection option for the RTT Control Block, J-Link RTT Viewer may not be able to find the `SEGGER_RTT` variable in RAM memory due to restrictions on memory access imposed by TrustZone settings made by the Renesas Device Partition Manager. Restrictions are typically applied for TrustZone Example Projects. If the RTT Control Block cannot be found by RTT Viewer, output from an Example Project may not be visible in the RTT Viewer Console with Auto Detection.



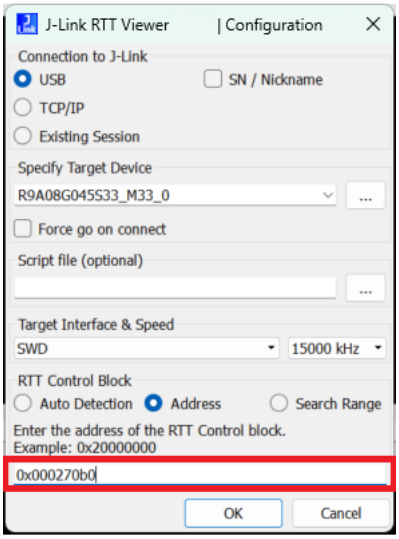
To mitigate this situation, you may use any one of the following approaches:

1. **Method 1:** The recommended approach is to search for _SEGGER_RTT variable in the map file, generated upon successfully building a configuration of an Example Project, which is by default located in the address space for On-chip SRAM.

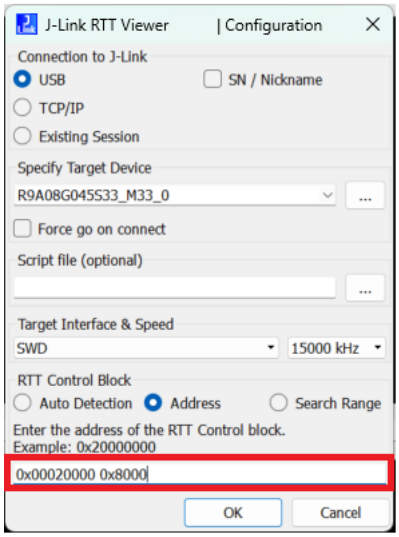



```
*(.bss*)
.bss._acDownBuffer
                0x00026ca0       0x10 ./src/SEGGER_RTT/SEGGER_RTT.o
.bss._acUpBuffer
                0x00026cb0      0x400 ./src/SEGGER_RTT/SEGGER_RTT.o
.bss._SEGGER_RTT
                0x000270b0       0xa8 ./src/SEGGER_RTT/SEGGER_RTT.o
                0x000270b0                _SEGGER_RTT
```

And input the exact address of the variable into the Address Input.



2. **Method 2:** Apply a search range within the first 32kB of SRAM Memory. This approach works when the compiler places _SEGGER_RTT variable in the first 32kB of SRAM memory.

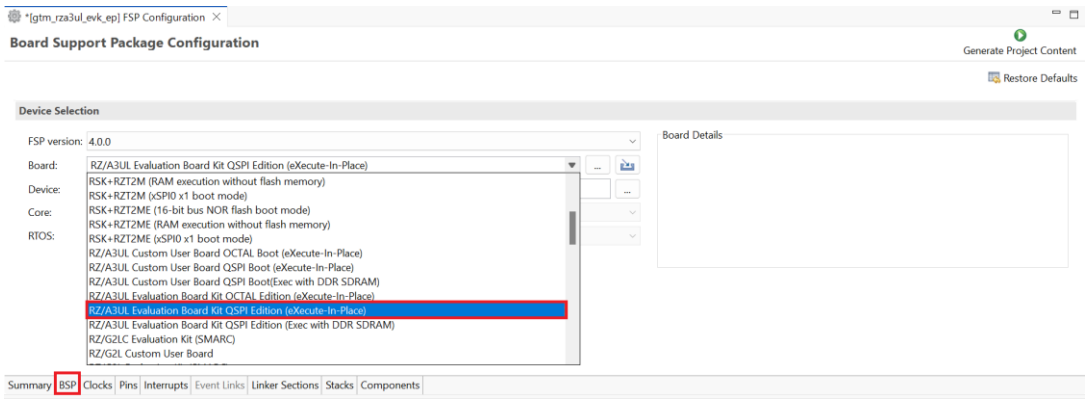


6.2 Change RZ/A3UL Board Edition Settings on the Project

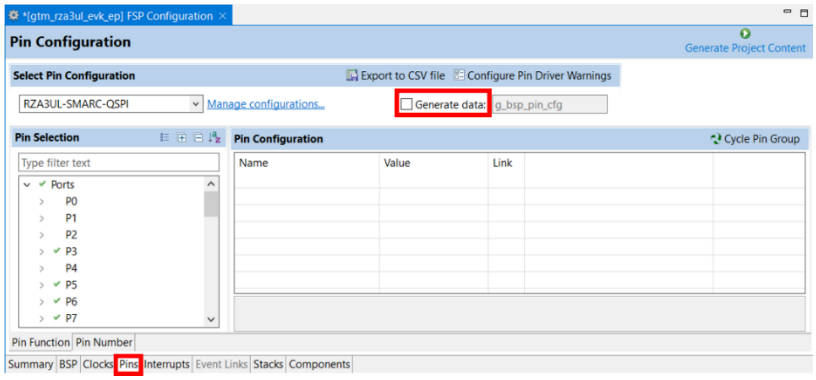
The example projects for RZ/A3UL set RZ/A3UL Evaluation Board Kit QSPI Edition (Exec with DDR-SDRAM) as default. If you can use the project for other board edition of RZ/A3UL, follow the procedure below.

6.2.1 Change to RZ/A3UL Evaluation Board Kit QSPI Edition (eXecute-In-Place)

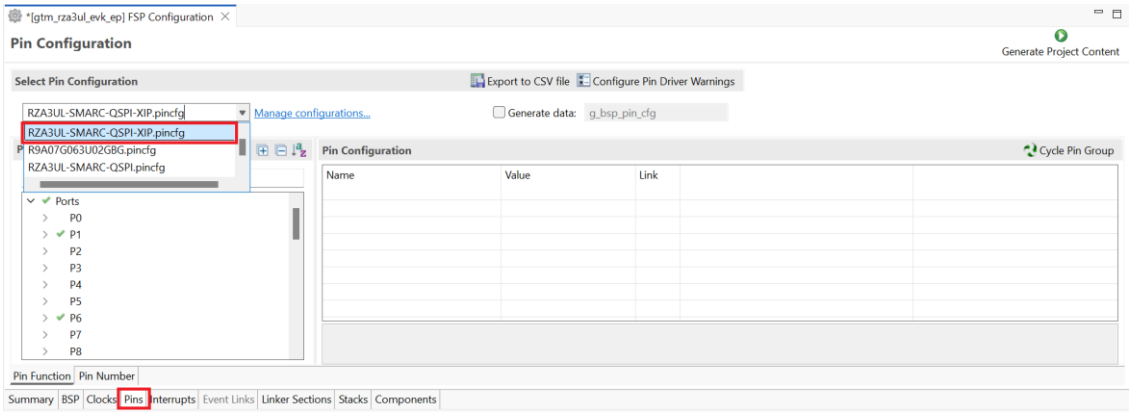
1. Change **Device Selection** to **RZ/A3UL Evaluation Board Kit QSPI Edition (eXecute-In-Place)** in FSP Configuration > BSP tab.



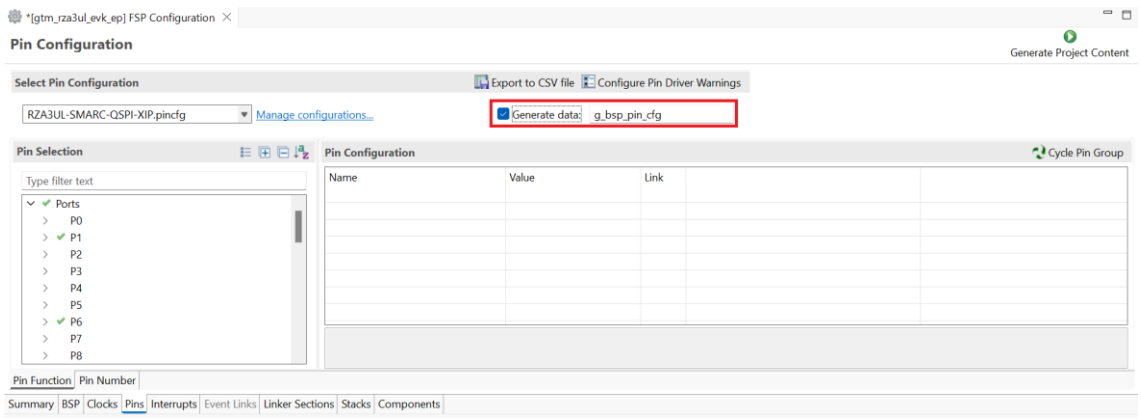
2. Uncheck the box in **Generate data** of FSP Configuration > Pins tab.



3. Change **Select Pin Configuration** to **RZA3UL-SMARC-QSPI-XIP.Pincfg** in FSP Configuration > Pins tab.



4. Check the box and enter **g_bsp_pin_cfg** in **Generate data** of FSP Configuration > Pins tab.

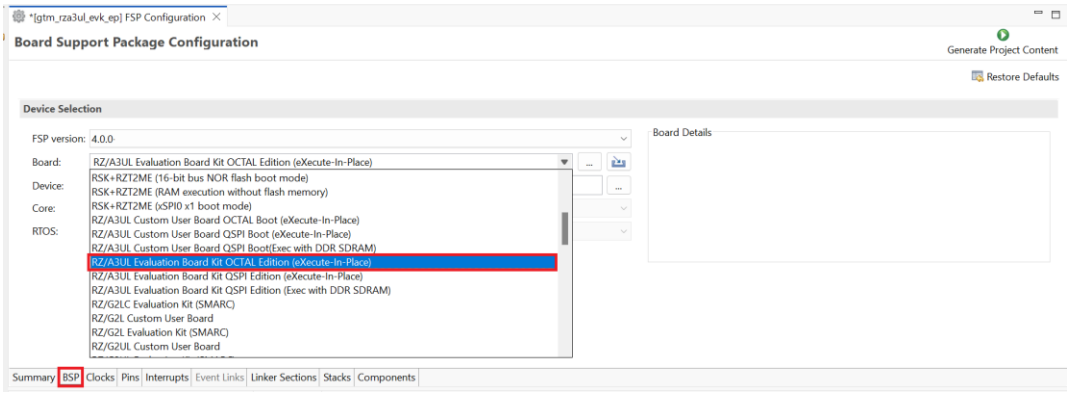


5. Click **Generate Project Content**.

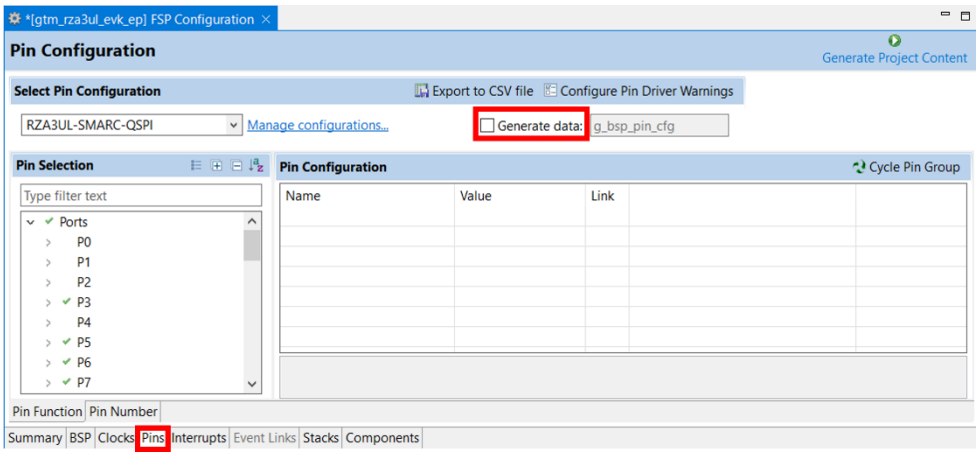
6. Build the project.

6.2.2 Change to RZ/A3UL Evaluation Board Kit OCTAL Edition (eXecute-In-Place)

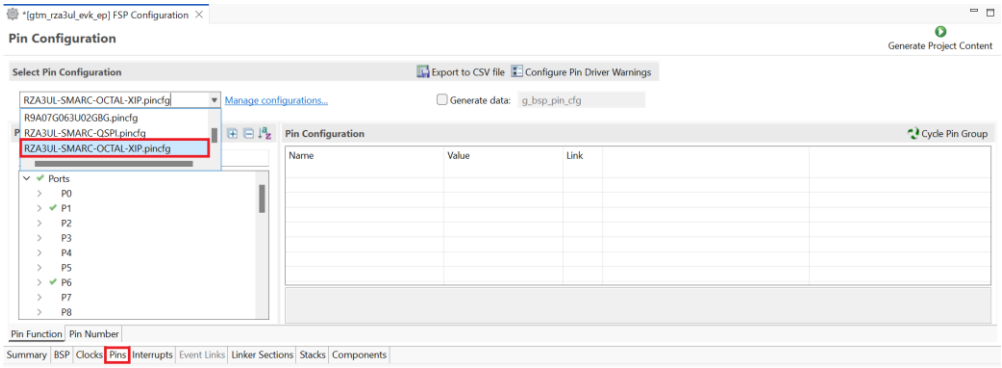
1. Change **Device Selection** to **RZ/A3UL Evaluation Board Kit OCTAL Edition (eXecute-In-Place)** in FSP Configuration > BSP tab.



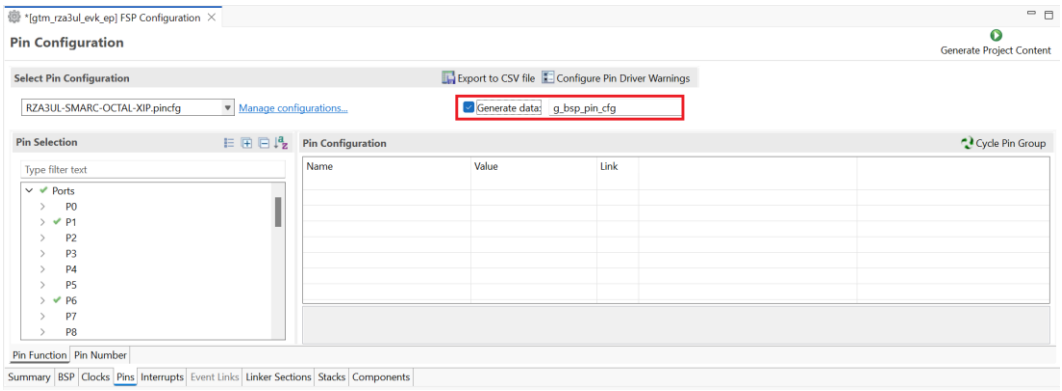
2. Uncheck the box in **Generate data** of FSP Configuration > Pins tab.



3. Change **Select Pin Configuration** to **RZA3UL-SMARC-OCTAL-XIP.pincfg** in FSP Configuration > Pins tab.

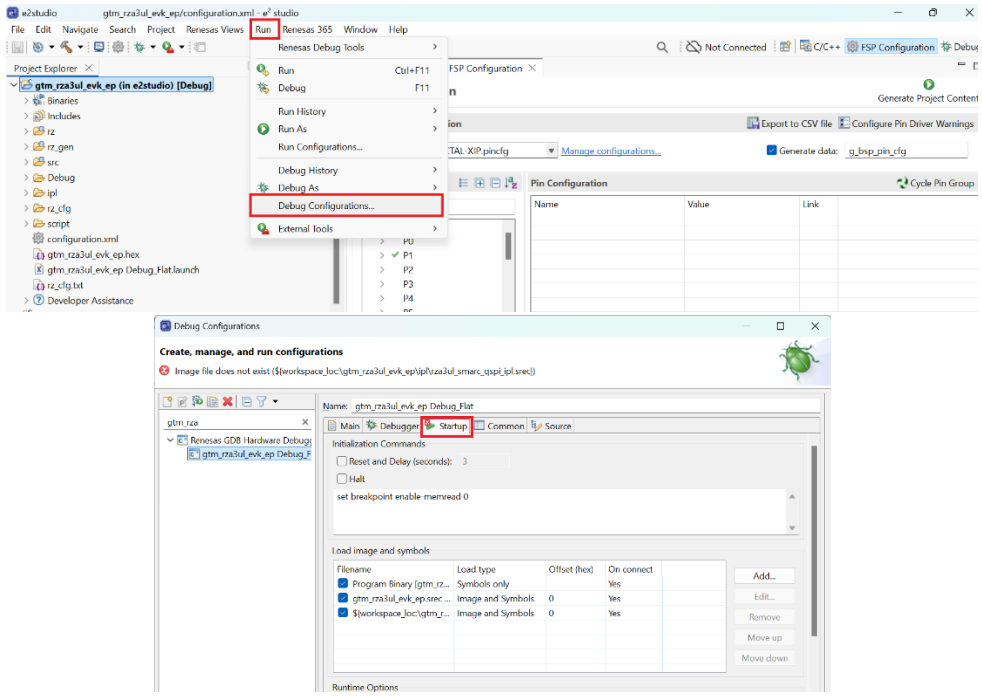


Check the box and enter **g_bsp_pin_cfg** in **Generate data** of FSP Configuration > Pins tab.

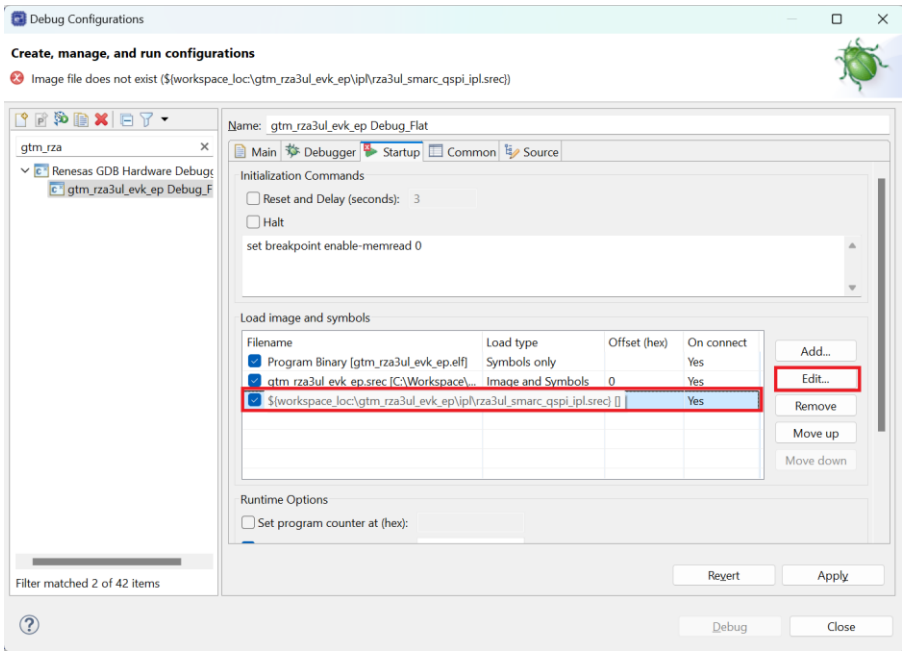


4. Click **Generate Project Content**.

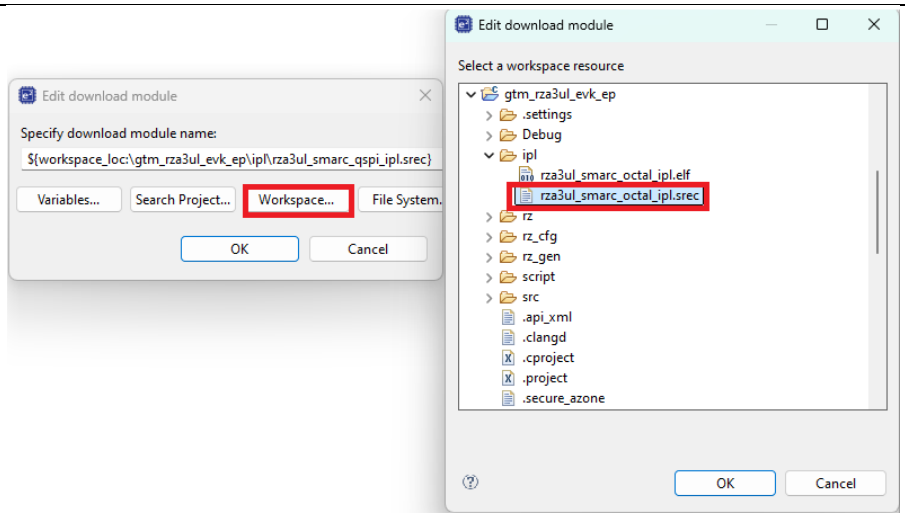
5. Build the project.
6. Change Loaded file of IPL for OCTAL edition.
   1. Open **Debug Configurations** and move to **Startup** tab.



2. Click rza3ul_smarc_qspi_ipl.srec in **Load Image and Symbols** and click **Edit…**.



3. Click **Workspace…** and select project/ipl/rza3ul_smarc_octal_ipl.srec.



7. Click OK and apply the Debug Configuration.

Revision History

Rev.	Date	Description	
		Page	Summary
1.00	Feb.24.26	—	Initial release
1.01	Mar.04.26	p5	Added section “4.2.1 Flashing pre-built binary”

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Publication Date: Mar.04.26

Published by: Renesas Electronics Corporation

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